



Project Report: Satellite Image Analysis for Deforestation Monitoring



Objective

The goal of this project is to build a system that detects and monitors deforestation in the Amazon rainforest using satellite imagery. The approach involves:

- Land classification using CNNs
- Change detection using sequential images
- Visualization of environmental changes over time



Dataset Description

Dataset:

Planet: Understanding the Amazon from Space (available via Kaggle)

- **Images:** Over 40,000 RGB satellite images (256x256 resolution)
- **Labels:** Each image is annotated with multiple tags (e.g., *primary forest*, *clear*, *agriculture*, *haze*)
- **Format:**
 - `/train-jpg/`: Directory with `.jpg` images
 - `train_v2.csv`: Contains `image_name` and associated land tags

Data Preprocessing Steps

1. Label Encoding:

- Each image's multi-label tags were converted into a **multi-hot encoded vector**.
- Tags were extracted, and a label dictionary was created to map each tag to an index.

2. Image Resizing:

- All images were resized to **128x128** for efficient training using CNNs.

3. Dataset & Dataloader Creation:

- A PyTorch **Dataset** class was implemented to load images and their labels.
- **Dataloader** was used for batching, shuffling, and feeding data into the CNN.

4. Train-Test Split:

- The dataset was split into 90% training and 10% validation sets using **train_test_split**.

Model Selection and Rationale

Model Used:

- A **Convolutional Neural Network (CNN)** was used for land classification.

Architecture Overview:

- 3 convolutional layers with ReLU + MaxPooling
- 2 fully connected layers
- Final output layer with **sigmoid activation** (multi-label output)

Rationale:

- CNNs are well-suited for image classification tasks.
- Using sigmoid instead of softmax allows **multi-label prediction**.
- The model is kept lightweight to reduce computational cost while still learning spatial features effectively.

Challenges Faced & Solutions

Challenge	Solution
Imbalanced Tags (e.g., many "primary")	Used weighted loss function and ensured diverse samples per batch
Large image sizes	Resized to 128x128 to reduce memory usage
Sequential Image Unavailability	Simulated change detection using random or duplicate samples for testing
Visualization rendering in notebooks	Embedded matplotlib-based dynamic image display functions

Outcome

- Successfully built a system that can classify land cover types from satellite imagery.
- Simulated change detection pipeline visualizes environmental changes over time.
- Notebook is modular and ready for fine-tuning with more complex models (e.g., ResNet, UNet).